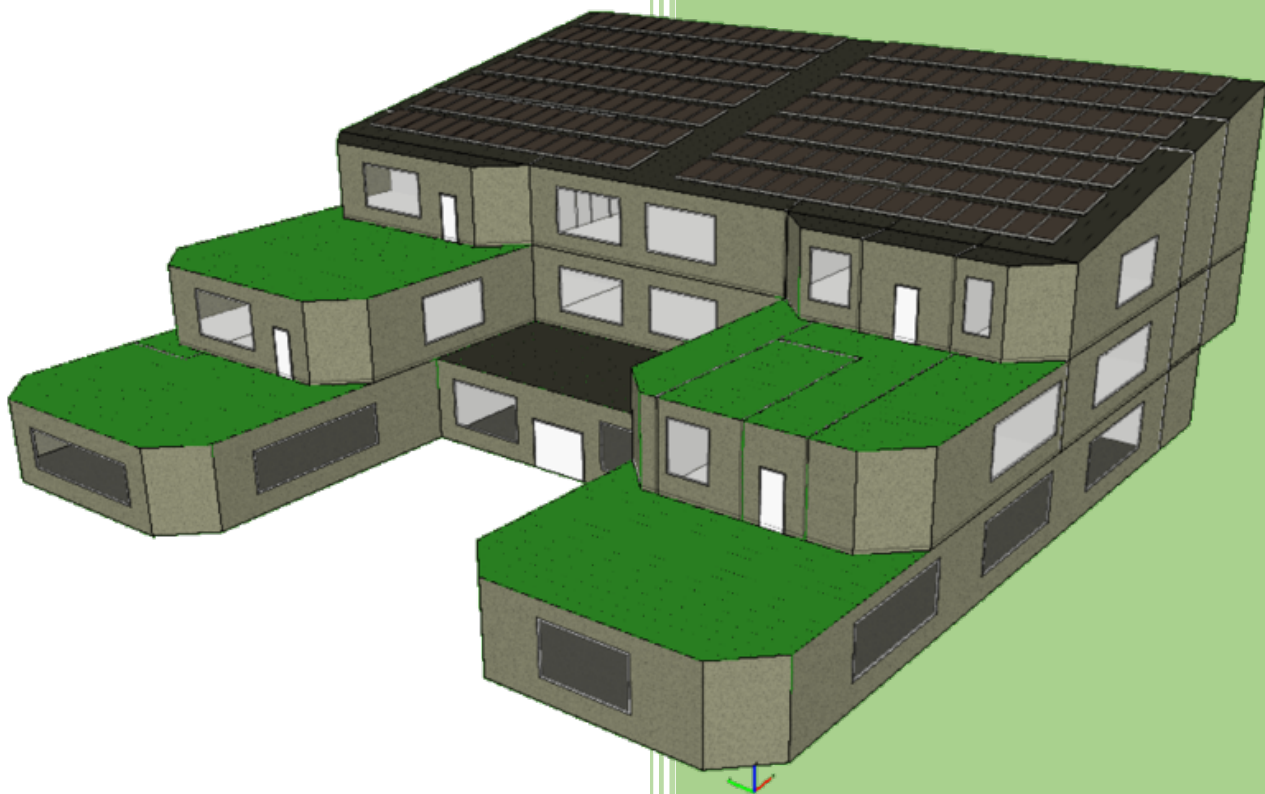


Group 7

Green Buildings Final Report



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Contents

Introduction	4
General Building Information	4
Location	4
Design	5
Energy consumption	6
LEED	9
Location & Transportation	9
Sensitive Land Protection	9
Surrounding density and diverse uses	9
Access to quality transit	10
Bicycle Facilities	11
Reduced Parking Footprint	11
Green Vehicles	11
Summary of Location & Transportation points	12
Sustainable Sites	12
Prerequisite: Construction activity pollution prevention	12
Site assessment	12
Site development - protect or restore habitat	12
Open Space	13
Rainwater management	13
Heat island reduction	13
Light pollution reduction	13
Summary of Sustainable Sites points	14
Water Efficiency	14
Prerequisite: Outdoor water use reduction	14
Prerequisite: Indoor water use reduction	14
Prerequisite: Building-level water metering	14
Outdoor water use reduction	15
Indoor water use reduction	15
Cooling tower water use	15
Water metering	16
Summary of Water Efficiency points	16
Energy & Atmosphere	16
Prerequisite: Fundamental commissioning and verification	16
Prerequisite: Minimum energy performance	16

Prerequisite: Building-level energy metering.....	16
Prerequisite: Fundamental refrigerant management	17
Enhanced commissioning	17
Optimize energy performance	17
Advanced energy metering.....	18
Demand response	18
Renewable energy production	18
Enhanced refrigerant management	18
Green power and carbon offsets	19
Summary of Energy & Atmosphere points.....	19
Materials & Resources	19
Prerequisite: Storage and collection of recyclables	19
Prerequisite: Construction and demolition waste management planning	19
Building life-cycle impact reduction.....	19
Building product - environmental product declarations	20
Building product - Sourcing of raw materials.....	20
Credit Building product - material ingredients.....	20
Construction and demolition waste management	20
Summary of Materials & Resources points.....	21
Indoor Environmental Quality.....	21
Prerequisite: Minimum Indoor Air Quality Performance.....	21
Prerequisite: Environmental Tobacco Smoke Control	21
Enhanced IAQ strategies	21
Low-emitting materials	21
Construction IAQ management plan	21
IAQ assessment.....	22
Thermal comfort	22
Interior lighting	23
Daylight	23
Quality views.....	23
Acoustic performance	23
Summary of Indoor Environmental Quality points	24
Innovation in Design	24
Innovation	24
LEED Accredited Professional	24
Summary of Innovation in Design points	24

Regional Priority	24
Regional Priority	24
Summary of Regional Priority points	25
LEED Points Summary & Certification	26
Discussion about LEED	26
Conclusion	27
References	28
Appendix.....	29
Appendix 1 - Water Consumption.....	29
Appendix 2 - Overall Energy Performance	31
Appendix 3 - NPV Analysis for PV system	32
Appendix 4 - Ventilation and IAQ Calculations	34
Appendix 5 - Fanger's Index.....	35
Appendix 6 - Daylight Simulation Results	39

Introduction

The purpose of this report is to design and evaluate a sustainable office building in Stockholm using the LEED v4 framework for New Construction and Major Renovations. The project aims to apply core green building principles to reduce environmental impact while ensuring compliance with Swedish regulations. The assessment focuses on quantifying improvements in energy performance, water efficiency, indoor environmental quality, and site-related sustainability. To support the analysis, the building is modelled and simulated in IDA ICE, to assess thermal performance, HVAC operation, and overall energy demand.

The report follows a clear structure: it outlines the building concept, evaluates the project against all relevant LEED v4 prerequisites and credits, and presents an energy analysis based on IDA ICE simulations and the Boverket's Building Regulations (BBR) baseline. The final sections briefly discuss the suitability of LEED in the Swedish context and summarize the expected certification level.

General Building Information

Location

The proposed office building is located on the KTH main campus in Stockholm, positioned between Brinellvägen and Teknikringen (Figure 1). The site lies within a densely built academic district characterized by educational buildings, research laboratories, and mixed green areas. The plot has a total area of approximately 3500 m², as specified in the project description. The specific site coordinates are 59°20'59"N 18°04'23"E.



Figure 1: 3D view of building location

The temperature profiles (Figure 2) confirm a predominantly cold climate, with long winters, frequent sub-zero conditions, and only short, mild summers. As a result, the building is heating-dominated, and most energy measures must prioritize winter performance.

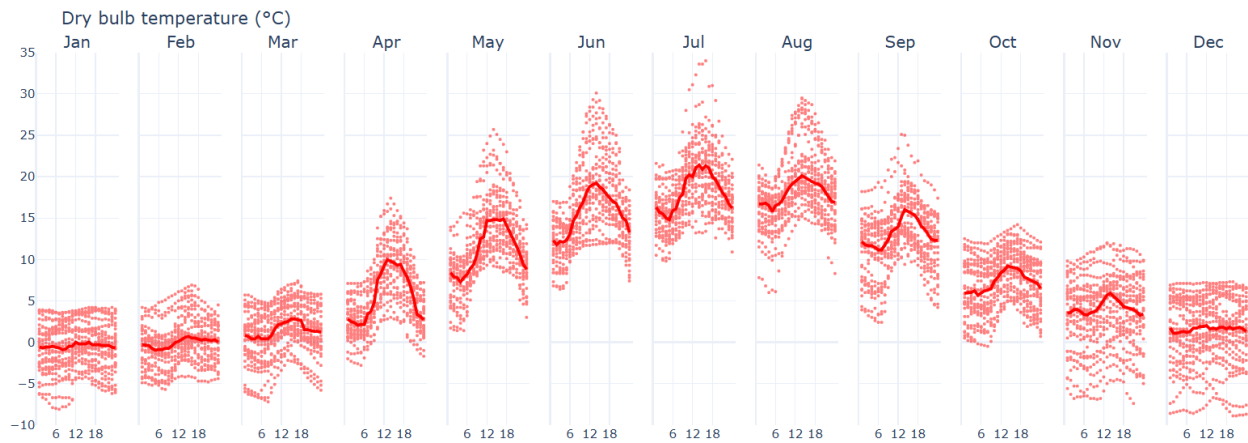


Figure 2: Daily Chart for dry bulb temperature distribution (CBE Clima Tool)

Design

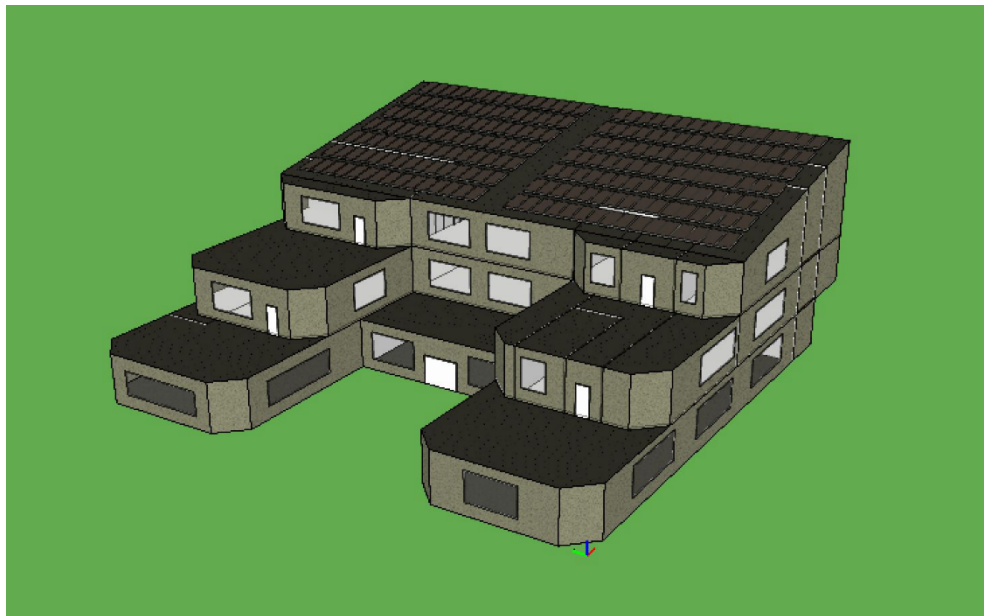


Figure 3: Building design, 3D view

The proposed building is a three-floors office facility with a total floor area of 3673 m² divided into 25 zones. The geometry is non-rectilinear and includes several protruding volumes on both the ground and upper floors. This configuration was maintained through the design phase to provide functional separation of spaces while remaining compatible with daylight-driven façade placement and circulation requirements.

The building is oriented towards the south. This orientation was selected to improve solar exposure on the roof, which is tilted to accommodate photovoltaic panels. The south-facing façades also allow for increased daylight availability compared to alternative orientations considered during the early modelling phase.

Windows and openings were positioned to obtain a reasonable distribution of natural light in perimeter spaces and to partially comply with LEED daylight criteria.

Table 1 summarizes the functional zoning of the building. The ground floor hosts the large common area and the gym, which occupy the central portion of the plan and serve as the main shared amenities. The managerial offices and conference rooms are concentrated on the south side of the building and the total area dedicated

to office functions (including open offices, private offices, and meeting rooms) spans the majority of the upper-floor perimeter.

Table 1: Summary of building zones and uses

Zone	Nr.	Tot. Surface [m ²]
Entrance/Acceptance	1	285
Main Offices	2	1405
Manager Offices	3	175
Conference Rooms	3	116
Break Rooms	2	357
Bathrooms	3	106
Gym	1	362
Changing Rooms	2	72
Common Area	1	455
Corridors	4	299
Elevators	3	36
Total	25	3673

Energy consumption

Proposed building's energy consumptions are related to the electricity used for lightning, equipment and HVAC (only ventilation) and energy utilized from the district network, both for heating and cooling.

A relevant distinction from operational (facility) and property (tenant) is fundamental to understand how the different consumptions impact in the total primary energy performances of the building (kWh/m²). Infact, operational electricity is not considered in the analysis of a building's Primary Energy Use according to BBR 2020, while property's is included. This difference substantially impacts the Primary Energy Use value.

Firstly, focusing on the tenant electricity and for a total of 142723 kWh, a 42% of that is used for the lightning, while the remaining 58% for the equipment needs. These electricity's consumptions are the ones for activities in premises, for lighting in offices, conference rooms and corridors and general offices machinery. A monthly distribution can be seen in Figure 4, where the olive green represents the lightning and the other green the equipment.

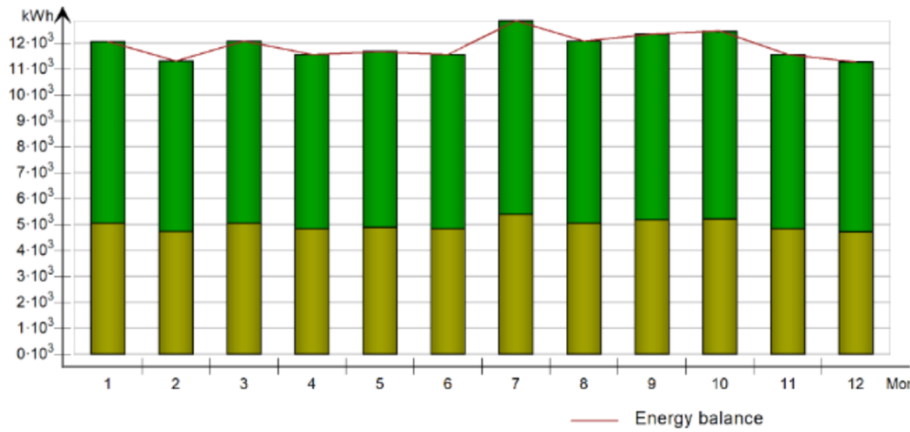


Figure 4: tenant's electricity consumptions

On the other hand, Figure 5 represents a monthly distribution of facility electricity usage, representing pumps, fans, elevators, cooling and lighting in common areas, stairwell and corridors. In the Figure, yellow is lightning, green equipment, light blue HVAC (ventilation system) and orange the PV's production.

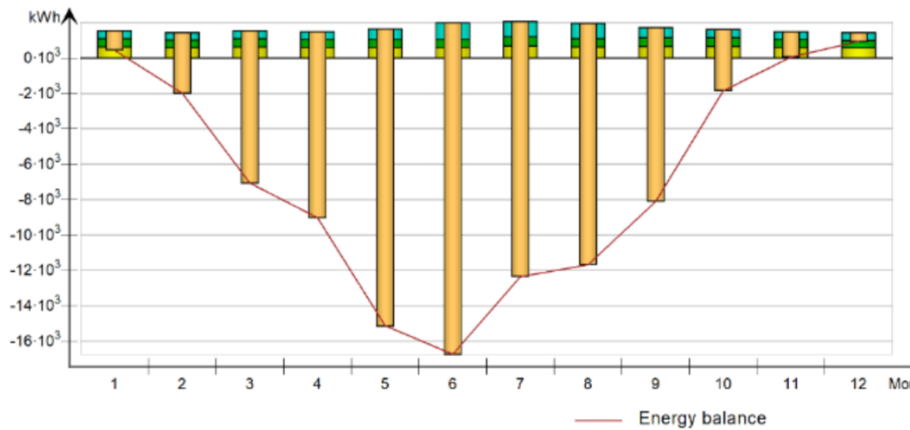


Figure 5: facility electricity usage

As can be seen, apart from January and December, the PVs always meet the facility consumptions and even produce a big amount of excess energy, meaning the system is over dimensioned. However, using the assumption of feed-in-tariff equal to the grid electricity's price, makes selling it to the grid profitable, getting closer in years the break-even date.

Generally, the demanded electricity by the operational utilities is much lower than the one required by the property. Moreover, a largely influencing aspect in the Primary Energy Use calculations is that both district heating and cooling are just included in the facilities consumptions. Figures 6 and 7 represent both of the trends during the year.

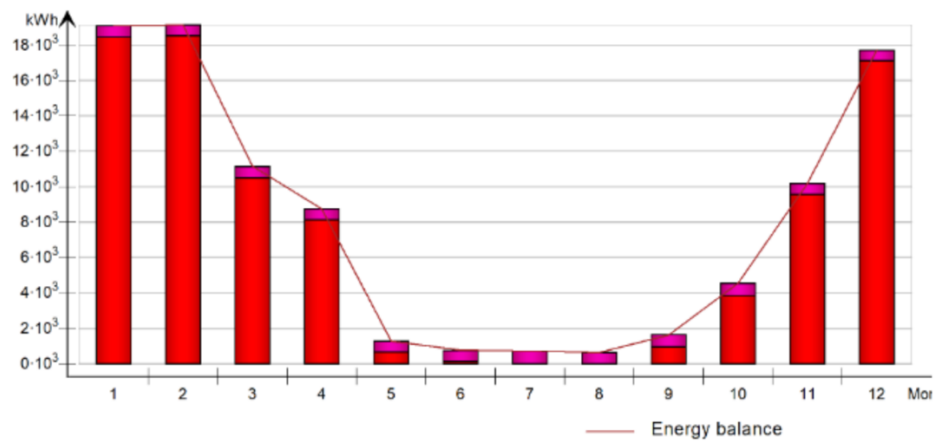


Figure 6: district heating demand

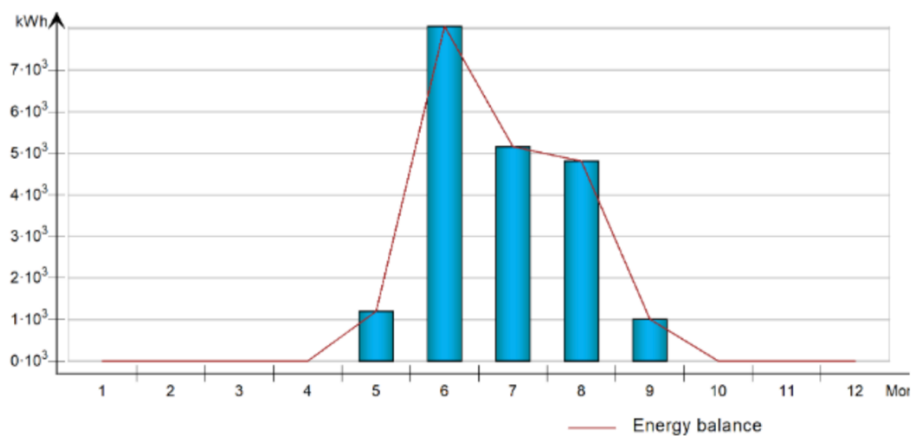


Figure 7: district cooling demand

District heating comprehends ambient (red) and DHW heating (purple). Both of the graphs reflect common Nordic hemisphere's trends, with warmer months between May and September and cooler ones, which as stated above have a bigger influence in consumptions in countries like Sweden, between October and April.

Overall, Table 2 resumes the energy consumptions of all building in kWh.

Table 2: general electricity consumptions

	Tenant	Facility	Total (Tenant + Facility)
Lightning	59802	7379	67181
Equipment	82921	5528	88449
HVAC (ventilation)	/	6883	6833
District Heating (including DHW)	/	95579	95579
District Cooling	/	20226	20226

LEED

This section presents the building's sustainability strategy according to the LEED v4 framework, outlining the credits for which points are targeted across the Energy and Atmosphere, Indoor Environmental Quality, Water Efficiency, Materials and Resources, and Sustainable Sites categories. For each credit, we summarize the design measures implemented, the expected performance outcomes, and the rationale behind credit selection. The section also includes an overview of the required investment costs, discussing how the proposed measures balance environmental benefits with economic feasibility.

Location & Transportation

Sensitive Land Protection

"To avoid the development of environmentally sensitive lands and reduce the environmental impact from the location of a building on a site."

To comply with the Sensitive Land Protection credit, the project assumes that the selected plot on the KTH campus qualifies as previously developed land, as it lies within an existing university area with no agricultural, ecological, or hydrological sensitivity. In line with LEED requirements, the project team would verify this condition by consulting municipal soil classifications, flood-risk maps, and available environmental records to confirm the absence of prime farmland, floodplains, protected habitats, wetlands, or water bodies within the development footprint. This fulfills the requirement for 1 point.

Surrounding density and diverse uses

"To conserve land and protect farmland and wildlife habitat by encouraging development in areas with existing infrastructure. To promote walkability, and transportation efficiency and reduce vehicle distance traveled. To improve public health by encouraging daily physical activity."

To demonstrate compliance with the LEED Diverse Uses credit, the project evaluates the availability of publicly accessible services and amenities located within an 800-meter walking distance from the proposed building entrance on the KTH campus. Using the geospatial analysis illustrated in Figure 8, multiple qualifying uses across several LEED categories have been identified, including retail (e.g., convenience store, kiosk, bookstore), services (e.g., hairdresser, café, restaurant, fitness center), civic and community facilities (e.g., library, university buildings), and public parks. In accordance with LEED counting rules, which allow no more than two uses per practice type and require representation from at least three of the five categories, the area contains more than eight eligible diverse uses within the allowed distance. Under these assumptions, the project is considered compliant with the credit requirements and would achieve the maximum allocation of 2 points.

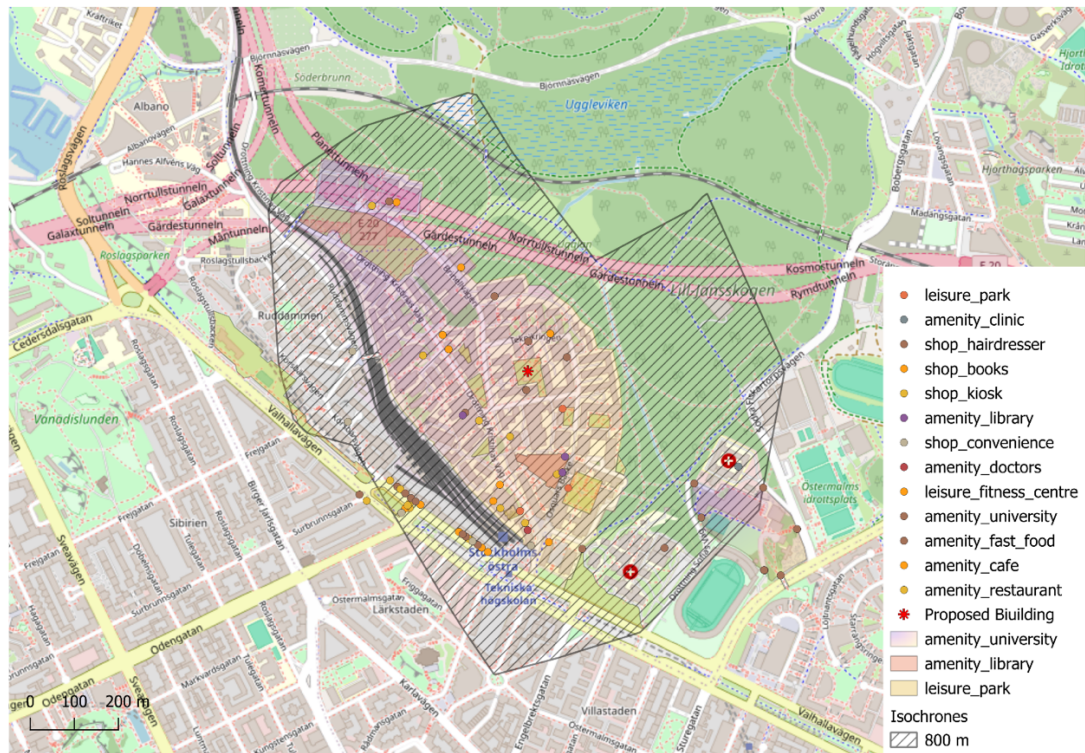


Figure 8: Map of the 800-meter walking area around the proposed building location (QGIS)

Access to quality transit

“To encourage development in locations shown to have multimodal transportation choices or otherwise reduced motor vehicle use, thereby reducing greenhouse gas emissions, air pollution, and other environmental and public health harms associated with motor vehicle use.”

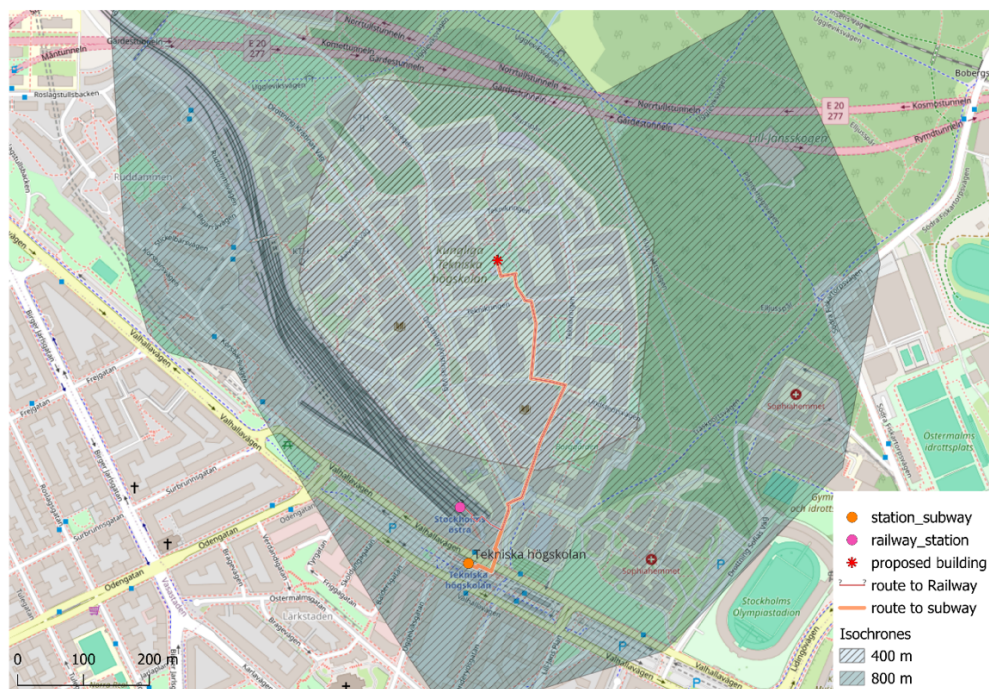


Figure 9: Walking-distance isochrones (400 m and 800 m) and pedestrian routes from the project site to nearby transit stations (QGIS)

According to the LEED requirements for Access to Quality Transit, the project is located within walking distance of two high-capacity transit services. The nearest metro station, Tekniska högskolan, is approximately 550 meters from the main entrance (Figure 9) and operates with a weekday frequency of one train every 5 minutes and weekend frequency of one train every 6 minutes, based on published schedules from Storstockholms Lokaltrafik (SL) [1]. In addition, the Stockholm Östra commuter rail station lies approximately 650 meters from the project site and provides frequent rail service throughout the day. The combined daily service from these two transit modes significantly exceeds the minimum trip thresholds established in Table 1 on page 90 of the LEED BD+C Reference Guide. This fulfills the requirement for 5 points.

Bicycle Facilities

“To promote bicycling and transportation efficiency and reduce vehicle distance traveled. To improve public health by encouraging utilitarian and recreational physical activity.”

The project meets the Bicycle Facilities credit by being located within 180 meters of the existing KTH campus bicycle network, which provides direct connections to multiple campus destinations and nearby transit stations. An outdoor covered bicycle storage area has been designated in the back of the building (Figure 10) to accommodate both long-term and short-term parking in accordance with LEED requirements, and on-site showers and changing facilities have been included to support bicycle commuting. This fulfills the requirement for 1 point.

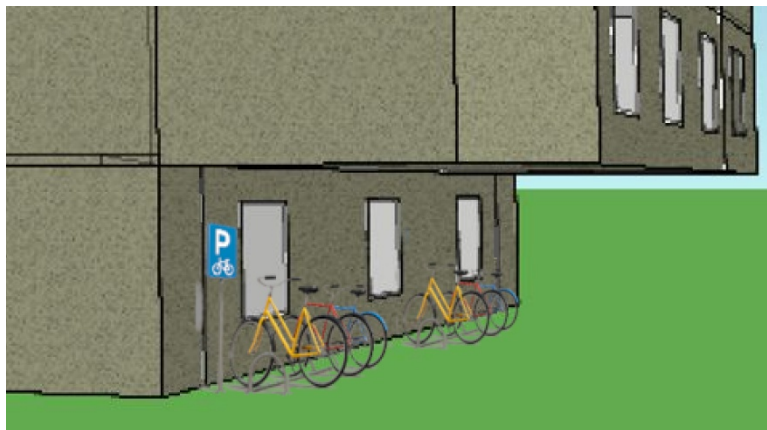


Figure 10: Outdoor bike parking space

Reduced Parking Footprint

“To minimize the environmental harms associated with parking facilities, including automobile dependence, land consumption, and rainwater runoff.”

The project is designed with a significantly reduced parking supply by relying on existing shared campus parking and not constructing any additional off-street parking beyond what is strictly necessary. The total number of spaces used by the project is assumed to remain below both the local code minimum and the base ratios from the ITE Transportation Planning Handbook, and at least 5% of these spaces will be designated as preferred parking for carpools. This fulfills the requirement for 1 point.

Green Vehicles

“To reduce pollution by promoting alternatives to conventionally fueled automobiles.”

The project will meet the Green Vehicles credit by assuming that 5% of the parking spaces used by building occupants can be designated as preferred parking for qualifying green vehicles within the existing campus parking supply. These spaces will be clearly marked, and it is further assumed that electric vehicle charging

equipment can be installed in 2% of the available spaces to satisfy the EVSE requirement. This fulfills the requirement for 1 point.

Summary of Location & Transportation points

Table 3: Location and Transportation points

LT Credit	Points Available	Points Achieved
LEED for Neighborhood Development Location	16	0
Sensitive Land Protection	1	1
High Priority Site	2	0
Surrounding Density and Diverse Uses	5	5
Access to Quality Transit	5	5
Bicycle Facilities	1	1
Reduced Parking Footprint	1	1
Green Vehicles	1	1
Total	16	14

Sustainable Sites

Prerequisite: Construction activity pollution prevention

“To reduce pollution from construction activities by controlling soil erosion, waterway sedimentation, and airborne dust.”

To satisfy the Construction Activity Pollution Prevention prerequisite, we assume that an erosion and sedimentation control plan will be prepared by a qualified professional in accordance with the Swedish regulatory framework and with the requirements of the EPA Construction General Permit, which LEED uses as a reference. The plan will be implemented and monitored throughout construction to prevent sediment runoff, dust generation, and disturbance of drainage systems. Under this assumption, the project is considered compliant with the prerequisite.

Site assessment

“To assess site conditions before design to evaluate sustainable options and inform related decisions about site design.”

A site assessment will be performed by a world-leading engineering consultancy, Sweco AB, which specializes in environmental site evaluation, local topography, hydrology, flood risk, existing vegetation and transportation infrastructure (Sweco) [2]. This assessment will be done focusing on the key design decisions to fulfill the requirements and achieve the credit in this category, performing building orientation assessment, stormwater management and landscape preservation strategies.

Site development - protect or restore habitat

“To conserve existing natural areas and restore damaged areas to provide habitat and promote biodiversity.”

The project minimized disturbance to only 730 m² at this previously greenfield site, with only 1,280 m² disturbed, including the 550 m² vegetated roof. This preserved the existing habitat of prairie grassland surrounding the building. Then 550 m² green roof will be composed by native/adapted Stockholm sedum species, thereby exceeding LEED requirements for habitat protection and restoration on greenfield sites, achieving full credit of 2 points.

Open Space

“To create exterior open space that encourages interaction with the environment, social interaction, passive recreation, and physical activities.”

Four accessible green roofs are present, for a total of 550 m² of accessible vegetated opens space. These spaces are offered for the occupants to have outdoor recreation, visual relief and connection to nature. These spaces enable the project to achieve the credit.

Rainwater management

“To reduce runoff volume and improve water quality by replicating the natural hydrology and water balance of the site, based on historical conditions and undeveloped ecosystems in the region.”

Following the previous paragraph, the green roof provides stormwater retention capacity. The real impact cannot be calculated, significantly reducing peak runoff rates and volumes compared to pure paved roofs. It can be assumed that one point out of the three available can be achieved.

Heat island reduction

“To minimize effects on microclimates and human and wildlife habitats by reducing heat islands.”

To calculate whether the project could meet the requirements in this category, the criterion present in Option 1 was analyzed:

$$\frac{\text{Area of Nonroof Measures}}{0.5} + \frac{\text{Area of High-Reflectance Roof}}{0.75} + \frac{\text{Area of Vegetated Roof}}{0.75} \geq \frac{\text{Total Site Paving Area}}{\text{Total Roof Area}}$$

Figure 11: LEED's Option 1 Heat Island reduction criterion

The total roof area accounts for 1550 m², but out of these 550 m² is covered by vegetative materials (green roof), 560 m² is covered by PV panels, 100 m² is a balcony, and 175 m² is used as corridors for workers when working with the PV plant (i.e., maintenance or cleaning). So, changing the parameters with the following values:

- Area of Nonroof Measures: 100 m² (balcony) + 175 m² (corridors) = 275 m²;
- Area of Vegetated Roof: 550 m²;
- Total Roof Area: 1550 m² - 560 m² (PV panels) = 990 m².

The left side of the equation (1283 m²) results to be greater than the right side (990 m²), enabling the project to achieve the 2 credits.

Light pollution reduction

“To increase night sky access, improve nighttime visibility, and reduce the consequences of development for wildlife and people.”

The outside lighting will not increase lighting pollution, since the low light lamps will be oriented to the ground, minimizing the upward light. This aligns with the light pollution reduction category and the single credit can be earned.

Summary of Sustainable Sites points

Table 4: Sustainable Sites points

SS Credit	Points Available	Points Achieved
Credit Site assessment	1	1
Credit Site development - protect or restore habitat	2	2
Credit Open space	1	1
Credit Rainwater management	3	1
Credit Heat island reduction	2	2
Credit Light pollution reduction	1	1
Total	10	8

Water Efficiency

In Appendix 1, the whole calculation process for both indoor and outdoor consumption is exposed.

Prerequisite: Outdoor water use reduction

“To reduce outdoor water consumption.”

Outdoor irrigation is limited to 550 m² vegetated roofs, which have a presence of a layer of dirt, grass and short native Stockholm species. Smart irrigation is used in the new construction, showing a reduction of water usage of 35%, passing from 659 L/day to 428 L/day, resulting in more than 30% reduced irrigation (following Option 2 of the prerequisite). Further analysis on the reduction will be done in the “Outdoor water use reduction” paragraph.

Prerequisite: Indoor water use reduction

“To reduce indoor water consumption.”

By 20% at least (41.45%)

Indoor water use is reduced by 41.15%, satisfying the 20% threshold required in the present prerequisite. Further details will be provided in the “Indoor water use reduction” section.

Prerequisite: Building-level water metering

“To support water management and identify opportunities for additional water savings by tracking water consumption.”

The building is equipped with a main drinkable water smart meter, that measures all the building’s water consumption at the point of connection to the main supply. The main meter records data real time and communicates them with a 15-min resolution, tracking detailed consumption trends. This configuration aligns then with the prerequisite.

Outdoor water use reduction

“To reduce outdoor water consumption.”

The baseline calculations for the outdoor water consumption were made thanks to the use of a LEED v5 Excel file, where the average monthly precipitation in Stockholm was used in input [3][4]. Thanks to the employment of smart irrigators (soil moisture sensors and rain sensors), a 35% outdoor water consumption savings can be assumed, following a study done in Northwestern Arkansas that showed an average reduction between 22.1% and 66.2% [5]. However, to achieve a point in the outdoor water use reduction category, a percentage of at least 50% was needed, so no points were assigned here.

Indoor water use reduction

“To reduce indoor water consumption.”

The baseline indoor water consumption accounts for 4,000 liters consumed per day, assuming that out of the 170 occupants, 90 are females and 80 males. According to LEED, it can be assumed that for toilet use, 3 flushes per female per day are made, while 1 WC flush and 2 urinals flushes per male per day. As for the showers present in the changing rooms connected with the gym, they will be used by gym attendees and bike riders. With these calculations, the consumption can be reduced to 2,342 L/day by implementing waterless urinals (bringing to zero the water usage in that category), and faucet adapters, both visible in Figure 12. The latter shows savings up to 70%, but a precautionary reduction factor of 50% was chosen instead.



Figure 12: example of a waterless urinal on the left (<https://www.cera-india.com/cera/sanitaryware-urinals/carys-waterless-urinal?srltid=AfmBOorUNqwQKBxN5laUy8cRTwt3hmOCP4rDRMHiuvCyyDb-gzCIQB4>), and faucet adapter on the right (<https://hygiene-shop.eu/en/faucet-adapter-contactless-water-saver-save-up-to-70-of-water-27>)

Ultimately, the indoor reduction accounts to 41.45% are reached, enabling the project to get 4 out of the 6 points available in this category.

Cooling tower water use

“To conserve water used for cooling tower makeup while controlling microbes, corrosion, and scale in the condenser water system.”

No cooling towers were employed, since the cooling system is provided by Stockholm’s district cooling. No points can be achieved in this category.

Water metering

“To support water management and identify opportunities for additional water savings by tracking water consumption.”

In addition to the main building water meter, the project includes sub-metering for major end uses and functional areas, such as the gym’s changing rooms and floor-by-floor distribution for the bathrooms. These meters are connected to the building management system with the data coming every 15-min, allowing clear and precise data analysis and verifying the performance of water-saving measures. The scope and resolution of this sub-metering strategy satisfy the requirements, enabling the project to achieve the credit.

Summary of Water Efficiency points

Table 5: Water Efficiency points

WE Credit	Points Available	Points Achieved
Credit Outdoor water use reduction	2	0
Credit Indoor water use reduction	6	4
Credit Cooling tower water use	2	0
Credit Water metering	1	1
Total	11	5

Energy & Atmosphere

Prerequisite: Fundamental commissioning and verification

“To support the design, construction, and eventual operation of a project that meets the owner’s project requirements for energy, water, indoor environmental quality, and durability.”

A formal commissioning process will be implemented throughout the design and construction phases, including development of a commission plan, performance verification testing of all building energy systems and production of a commissioning report documenting that such systems operate as designed. This prerequisite fulfills LEED v4 requirements for fundamental commissioning and verification.

Prerequisite: Minimum energy performance

“To reduce the environmental and economic harms of excessive energy use by achieving a minimum level of energy efficiency for the building and its systems.”

Energy modeling simulations using IDA ICE demonstrate a 46% cost reduction in the proposed building’s total annual energy consumption compared to the baseline building model (calculated per BBR 2020 Swedish standards). This significant improvement in energy performance rating substantially exceeds the mandatory 5% improvement threshold required by LEED v4, ensuring compliance with the Minimum Energy Performance prerequisite.

Prerequisite: Building-level energy metering

“To support energy management and identify opportunities for additional energy savings by tracking building-level energy use.”

The building will be equipped with comprehensive real-time energy metering system, including a main meter for total consumption monitoring and individual sub-meters tracking district heating, district cooling, and electrical consumption separately. Smart meter technology with continuous data acquisition will enable real-time performance monitoring and supports future integration with grid demand response programs, fulfilling the Building-Level Energy Metering prerequisite.

Prerequisite: Fundamental refrigerant management

“To reduce stratospheric ozone depletion.”

The building's cooling demands are met entirely through Stockholm's district cooling network, eliminating the need for on-site refrigeration systems. Stockholm's district cooling infrastructure is supplied by Vattenfall/Fortum and operates with low-GWP refrigerants (primarily ammonia-based systems with GWP < 50) compliant with EU F-Gas Regulation (EU 517/2014) and LEED v4 refrigerant management requirements. This eliminates refrigerant-related environmental impacts and fulfills the Fundamental Refrigerant Management prerequisite [6].

Enhanced commissioning

“To further support the design, construction, and eventual operation of a project that meets the owner's project requirements for energy, water, indoor environmental quality, and durability.”

An Enhanced Commissioning process is assumed to be implemented to ensure optimal building performance with the aim of identifying all critical energy systems. Comprehensive testing and verification procedures are scheduled during the construction phase to confirm that all systems operate in accordance with design specifications and acceptance criteria. Following building occupancy, a post-occupancy commissioning phase (first 10 months) will be conducted to identify and correct any operational issues, optimize system performance, and ensure that actual energy consumption aligns with design projections. Building occupants and maintenance staff will receive training on building systems operations and systems manual documenting will be provided for long-term operational guidance. This approach to enhanced commissioning achieves 2 points and establishes a foundation for sustained energy performance throughout the building's operational life.

Optimize energy performance

“To achieve increasing levels of energy performance beyond the prerequisite standard to reduce environmental and economic harms associated with excessive energy use.”

The most impactful category of the LEED certification has been handled by adding PV panels on top of the main roof, without any additional tilt angle, since the whole concept of the roof is to be tilted and south faced. To be exact, 217 solar panels were added, each one with a rated power of 530 Wp. The rows were distanced between each other to facilitate the work of the personnel when needed, plus, two corridors on the extremes and a big one in the middle was added, dividing the whole plant into two separate arrays.

To calculate the savings in the yearly cost of energy, the following prices were used:

- 1.5 SEK/kWh for the electricity price;
- 0.85 SEK/kWh for the district heating;
- 0.50 SEK/kWh for the district cooling.

And to further calculate the revenues coming from the export of the surplus energy generated:

- 1.5 SEK/kWh as feed-in price.

As can be seen in the overall energy performances in Appendix 2 of the Proposed Building, the electricity exported is very high, which, with the assumption made, gave the project a 45.8% reduction in yearly energy price, accounting for 16 of the 18 available points in this category. This value is taken from the sum of the yearly costs (considering the exports as negative), passing from the total cost of the baseline building of 335,552 SEK to 181,757 SEK per year.

For the sake of the project and the economic feasibility of the investment, an NPV cost analysis (Appendix 3) was made for the PV panels, showing a promising break-even point at year eight after the initial investment, assuming a specific investment cost of 8,500 SEK/kW installed [7].

Advanced energy metering

“To support energy management and identify opportunities for additional energy savings by tracking building-level and system-level energy use”

Advanced energy metering will be implemented with sub-meters installed for the HVAC system, enabling separate tracking of facility electricity (common areas, mechanical systems) versus tenant electricity (individual office spaces and private areas). Additionally, data from sub-meters and the main smart meter will be recorded at 15-minute intervals. This energy consumption visibility meets and exceeds LEED v4 requirements achieving 1 point for Advanced Energy Metering.

Demand response

“To increase participation in demand response technologies and programs that make energy generation and distribution systems more efficient, increase grid reliability, and reduce greenhouse gas emissions.”

Demand response capability will be enabled through integration with Stockholm’s grid operator (Vattenfall). Smart meters will provide 15-minute real-time data, while the main meter and HVAC sub-meters allow identification and automated control of flexible loads through the BMS. This infrastructure supports participation in utility demand response programs, with the building owner committed to reducing peak loads during high-demand periods. These measures achieve 1 point for Demand Response; 2 points would be attainable with automated load shedding during utility-initiated events.

Renewable energy production

“To reduce the environmental and economic harms associated with fossil fuel energy by increasing self-supply of renewable energy.”

A 217 kWp photovoltaic (PV) solar array has been installed on the building's roof, generating approximately 102.2 MWh of renewable electricity annually. Compared to the building's total annual energy consumption of 162.5 MWh, the on-site PV system covers 62.8% of the building's operational energy needs, demonstrating substantial on-site renewable energy generation. This exceeds LEED v4 thresholds for maximum renewable energy credit eligibility, achieving 3 points for Renewable Energy Production.

Enhanced refrigerant management

“To reduce ozone depletion and support early compliance with the Montreal Protocol while minimizing direct contributions to climate change.”

Enhanced refrigerant management is achieved using district cooling, eliminating the need for on-site refrigerant-based chillers and ensuring zero ozone-depleting substances and no direct refrigerant emissions. This strategy achieves 1 point for Enhanced Refrigerant Management.

Green power and carbon offsets

“To encourage the reduction of greenhouse gas emissions through the use of grid-source, renewable energy technologies and carbon mitigation projects.”

The building owner has committed to a green power procurement strategy whereby electricity supplied to the building is sourced from energy providers meeting the LEED v4 requirement of minimum 50% renewable energy generation. This commitment covers a minimum operational period of 2 years, as required by LEED standards. Green power certification through *Ekocentrum* (Swedish Green Electricity Label) or equivalent will guarantee that at least half of the building's electricity consumption comes from certified renewable sources. This commitment to green power procurement achieves 2 points for Green Power and Carbon Offsets.

Summary of Energy & Atmosphere points

Table 6: Energy & Atmosphere points

EA Credit	Points Available	Points Achieved
Credit Enhanced commissioning	6	2
Credit Optimize energy performance	18	16
Credit Advanced energy metering	1	1
Credit Demand response	2	1
Credit Renewable energy production	3	3
Credit Enhanced refrigerant management	1	1
Credit Green power and carbon offsets	2	2
Total	33	26

Materials & Resources

Prerequisite: Storage and collection of recyclables

“To reduce the waste that is generated by building occupants and hauled to and disposed of in landfills.”

The project provides integrated, dedicated areas accessible for collection and storage of mixed paper, cardboard, glass, plastics, and metals, along with proper provisions for safe handling of batteries and electronic waste. This fully complies with the LEED v4 prerequisite.

Prerequisite: Construction and demolition waste management planning

“To reduce construction and demolition waste disposed of in landfills and incineration facilities by recovering, reusing, and recycling materials.”

A company will be hired to develop and implement a Construction and Demolition Waste Management Plan that targets diversion of at least five material streams, specifies diversion methods, and tracks waste and recycling rates, fulfilling the LEED v4 prerequisite.

Building life-cycle impact reduction

“To encourage adaptive reuse and optimize the environmental performance of products and materials.”

Within the final design strategy, a portion of the building's enclosure and interior non-structural elements will be sourced from reused or salvaged materials from certified off-site suppliers such as *Fabege Reuse Hub* [8]. In particular, the project plans to reuse approximately 25-30% of the total project surface area, including

selected facade cladding components, internal partition systems, and portions of the raised flooring, securing 2 points.

Building product - environmental product declarations

“To encourage the use of products and materials for which life-cycle information is available and that have environmentally, economically, and socially preferable life-cycle impacts. To reward project teams for selecting products from manufacturers who have verified improved environmental life-cycle impacts.”

The project will use multiple permanently installed products from various manufacturers with Environmental Product Declarations that meet LEED v4 disclosure requirements. While detailed verification of life-cycle impact reduction plans is not available, standard industry practice in material selection supports eligibility to earn 1 point for Building Product Disclosure and Optimization under LEED v4.

Building product - Sourcing of raw materials

“To encourage the use of products and materials for which life cycle information is available and that have environmentally, economically, and socially preferable life cycle impacts. To reward project teams for selecting products verified to have been extracted or sourced in a responsible manner.”

We plan to specify at least 20 different permanently installed products from a minimum of five manufacturers who will be requested to ensure their materials comply with LEED ingredient disclosure requirements. This assumption supports potential eligibility for 2 points under the Material Ingredients credit, contingent on manufacturer cooperation and documentation.

Credit Building product - material ingredients

“To encourage the use of products and materials for which life-cycle information is available and that have environmentally, economically, and socially preferable life-cycle impacts. To reward project teams for selecting products for which the chemical ingredients in the product are inventoried using an accepted methodology and for selecting products verified to minimize the use and generation of harmful substances. To reward raw material manufacturers who produce products verified to have improved life-cycle impacts.”

We plan to specify a broad range of permanently installed products with verified life-cycle information and reduced-impact profiles. At least 20 products from five manufacturers will be selected with documented environmental and health attributes, meeting accepted disclosure and optimization pathways. Based on these assumptions and expected manufacturer compliance, the project is positioned to achieve 2 points under the Material Ingredients credit.

Construction and demolition waste management

“To reduce construction and demolition waste disposed of in landfills and incineration facilities by recovering, reusing, and recycling materials.”

The project will involve *Sortera AB* [9], a leading waste management company in Stockholm. Their proximity to KTH and expertise support achieving 2 points under the LEED v4 Construction and Demolition Waste Management credit by diverting at least 75% of waste across four or more material streams.

Summary of Materials & Resources points

Table 7: Materials & Resources points

MR Credit	Points Available	Points Achieved
Credit Building life-cycle impact reduction	5	2
Credit Building product - environmental product declarations	2	1
Credit Building product - sourcing of raw materials	2	2
Credit Building product - material ingredients	2	2
Credit Construction and demolition waste management	2	2
Total	13	9

Indoor Environmental Quality

Prerequisite: Minimum Indoor Air Quality Performance

“To contribute to the comfort and well-being of building occupants by establishing minimum standards for indoor air quality (IAQ).”

The project meets the prerequisite through mixed-mode ventilation with central AHU heat recovery and VAV systems designed following the ASHRAE Ventilation Rate Procedure, delivering a total of 3,858 m³/h total outdoor air across 1,405 m² offices, see Appendix 4 for the calculation. VAV outdoor airflow with high accuracy and with alarms at 15% deviation from the 3,858 m³/h per zone setpoints, supported by CO₂ placed at 3m height on the walls, alerting at 10% above setpoints. Naturally ventilated offices incorporate operable windows designed for Stockholm climate performance, and filtration on all mechanical supply air. Building pressurization maintains positive pressure preventing infiltration.

Prerequisite: Environmental Tobacco Smoke Control

“To prevent or minimize exposure of building occupants, indoor surfaces, and ventilation air distribution systems to environmental tobacco smoke.”

Smoking is prohibited in all indoor areas; outdoor smoking spaces are designated more than 7.5 meters from entrances and air intakes, in full compliance with ETS control requirements.

Enhanced IAQ strategies

“To promote occupants’ comfort, well-being, and productivity by improving indoor air quality.”

An HVAC system with external air filtration and active air recirculation will be installed, ensuring improved indoor air quality compared to regulatory minimums.

Low-emitting materials

“To reduce concentrations of chemical contaminants that can damage air quality, human health, productivity, and the environment.”

All interior finishing materials meet low VOC (Volatile Organic Compound) emission requirements, minimizing indoor air pollutants and supporting occupant health, achieving the credits.

Construction IAQ management plan

“To promote the well-being of construction workers and building occupants by minimizing indoor air quality problems associated with construction and renovation.”

A Construction Indoor Air Quality Management Plan will be implemented throughout the building phase in compliance with LEED v4 requirements. The plan includes systematic dust control measures, protection of absorptive materials from moisture damage, continuous site ventilation during construction, and documentation of all implemented procedures. The management and supervision of these measures will be delegated to *Noxer AB*, a specialized environmental management company based in Stockholm, which provided regular site inspections and compliance documentation to ensure adherence to the IAQ management protocol.

IAQ assessment

“To establish better quality indoor air in the building after construction and during occupancy.”

An Indoor Air Quality Assessment will be conducted prior to building occupancy through a three-day 100% outdoor air flush-out procedure, as required by LEED v4 standards. The procedure can be executed and certified by *Sweco AB* (Environmental Division, Stockholm office), an indoor environmental quality testing firm, which will also provide the necessary documentation and commissioning records to confirm whether the IAQ requirements are met.

Thermal comfort

“To promote occupants’ productivity, comfort, and well-being by providing quality thermal comfort.”

A thermal comfort for new buildings, like the one studied in this project, can consider a comfort acceptability range when the predicted mean vote (PMV) is between -0.5 and 0.5, and a predicted percentage dissatisfied (PPD) less than 10%, following the European regulation ISO 7730-2005 category B (Figure 13).

Category		Description	Allowable predicted mean vote	Allowable predicted percentage dissatisfied
ISO 7730- 2005	EN 15251- 2007			
A	I	Recommended for spaces occupied by very sensitive and fragile persons with special requirements (very young children, elderly, ill)	$-0.2 < PMV < 0.2$	< 6%
B	II	Suitable for most new buildings and renovations	$-0.5 < PMV < 0.5$	<10%
C	III	Suitable for existing buildings	$-0.7 < PMV < 0.7$	<15%
	IV	Values other than above; acceptable for only part of year	$PMV < -0.7$ or $PMV > 0.7$	>15%

Figure 13: European regulation thermal comfort guidelines

Since showing all the recap tables for all the months and all the occupied area is repetitive and extremely extensive, in Appendix 5 the Fanger’s indexes of two key zones in July and December are shown, the Office First and the Gym. The first has been chosen since it’s the largest zone and shows similar numbers throughout the year to all the other occupied zones, while the gym has been shown since it’s the only zone that has most of the hours out of the comfort acceptability range. The months selected, instead, represent the two extremes during the year, being July the one that shows the highest values of PPD and PMV, while December the lowest (on average).

As for the Office First (and, consequently, all the other frequently occupied zones, such as the other offices, conference rooms, cafeterias, entrance and acceptance, and common area), PPD and PMV is always inside the range throughout the year, apart from few peaks of heat during the hot summer hours. While as for the Gym, which is by far the worst zone, the thermal comfort is inside the range just few times because of the type of activity that has been done there.

Ultimately, since all the occupied areas (apart from the Gym), have Fanger's values inside the range during the year, the single credit point in this category is considered achieved.

Interior lighting

"To promote occupants' productivity, comfort, and well-being by providing high-quality lighting."

Users are allowed to adjust illumination levels according to their needs, assuming that at least 50% of the regularly occupied spaces have the controls to do so.

Daylight

"To connect building occupants with the outdoors, reinforce circadian rhythms, and reduce the use of electrical lighting by introducing daylight into the space."

Daylight compliance was evaluated using Option 2 (simulation-based illuminance calculations). Illuminance levels were computed at 9:00 and 15:00 on the equinox for all regularly occupied spaces, following the LEED methodology. As shown in Appendix 6, 40 out of 52 measured points fall within the required 300-3000 lux range for both time steps, corresponding to 77% of the regularly occupied floor area meeting the daylight illuminance criteria. In general, three-pane windows were used in the building, having a good lighting transmittance. For the few areas that were showing too much illuminance, windows with worse lighting transmittance were chosen (four-pane windows). In those areas where the natural lighting was too low, instead, windows with better lighting transmittance were taken, at the cost of increasing the final U-value in the building.

The percentage exceeds the 75% threshold defined in Table 2 of the LEED BD+C Reference Guide for Option 2 and fulfills the requirement for 1 point.

Quality views

"To give building occupants a connection to the natural outdoor environment by providing quality views."

The project achieves the Quality Views credit by providing direct line of sight views to the outdoors for 75% of all regularly occupied spaces, with views extending at least 25 feet (7.5 meters) beyond the glazing to natural elements including the green roofs. To enhance this, most of the occupied zones have windows on different orientation, while the other zones usually feature a big window only on one side.

Acoustic performance

"To provide workspaces and classrooms that promote occupants' well-being, productivity, and communications through effective acoustic design."

No formal acoustic study was conducted; however, most of the exterior windows feature triple glazing, which reduces facade noise penetration. Not enough to get the LEED point.

Summary of Indoor Environmental Quality points

Table 8: Indoor Environmental Quality points

Eq Credit	Points Available	Points Achieved
Credit Enhanced IAQ strategies	2	2
Credit Low-emitting materials	3	3
Credit Construction IAQ management plan	1	1
Credit IAQ assessment	2	1
Credit Thermal comfort	1	1
Credit Interior lighting	2	2
Credit Daylight	3	1
Credit Quality views	1	1
Credit Acoustic performance	1	0
Total	16	12

Innovation in Design

Innovation

“To encourage projects to achieve exceptional or innovative performance.”

These credits are not achieved with the proposed building. In fact, even if the construction has good overall performance, it doesn't achieve noteworthy and exemplary environmental performance using a strategy not addressed in the LEED green building rating system.

LEED Accredited Professional

“To encourage the team integration required by a LEED project and to streamline the application and certification process.”

None of the members of the project is a LEED accredited professional.

Summary of Innovation in Design points

Table 9: Materials & Resources points

Eq Credit	Points Available	Points Achieved
Credit Innovation	5	0
Credit LEED Accredited Professional	1	0
Total	6	0

Regional Priority

Regional Priority

“To provide an incentive for the achievement of credits that address geographically specific environmental, social equity, and public health priorities.”

Regional Priority credits satisfaction was established by looking at the criteria for humid continental and urban areas in Sweden, where Stockholm is located. The maximum achievable points are four.

Table 10: Regional Priority subcategories points thresholds

Regional Priority Credit	Point Threshold
Optimize Energy Performance	12 points
Site Development - Protect or Restore Habitat	2 points
Indoor Water Use Reduction	1 point
Building Life Cycle Impact Reduction	2 points
Building Products Disclosure and Optimization - Material Ingredients	2 points
Quality Views	1 point

Considering the thresholds for New Constructions shown in the Table above and the already achieved points, the four points for this section will be accomplished reaching the threshold of “Optimize Energy Performance”, “Quality Views”, “Indoor Water Use Reduction” and “Building Products Disclosure and Optimization - Material Ingredients”.

Summary of Regional Priority points

Table 11: Regional Priority achieved points

Eq Credit	Points Available	Points Achieved
Credit Regional Priority	4	4
Total	4	4

LEED Points Summary & Certification

Table 12 below shows the summary of all the points achieved in all the categories.

Table 12: Summary of all points achievable and achieved per category

Topic Area	Possible Points	Points Achieved
Location & Transportation	16	14
Sustainable Sites	10	8
Water Efficiency	11	5
Energy & Atmosphere	33	26
Materials & Resources	13	9
Indoor Environmental Quality	16	12
Innovation in Design	6	0
Regional Priority	4	4
Total	109	78

The final score is 78, enabling the building to achieve a gold certification level.

Discussion about LEED

LEED v4 provides a clear framework for evaluating the sustainability of a building, but its direct application in Sweden faces some limitations that affect the robustness of the result. Swedish building regulations (i.e. BBR and local energy standards), impose already stringent requirements on envelope performance, ventilation and, mostly, district heating connections, that in some cases LEED over-rewards. A more appropriate assessment method, tailored to the Swedish scenarios, such as Miljöbyggnad, would likely have provided a more accurate reflection of the project's actual sustainability performance

For this specific project, the final LEED score should not be interpreted as proof that the building is highly sustainable in absolute terms. Many decisions were made to achieve points, such as employing worse windows to get the illuminance credit. The design process focuses early on in-loco site energy production, rather than first minimizing demand and improving passive performance. For this matter, important opportunities to reduce heating, cooling and electrical demand at source were only partially explored. In addition, the financial and environmental results are strongly influenced by simplified assumption, for example the exported electricity is valued as the same price as the imported one, when it's well known that the electricity price is lower when the PV production is globally high. Another proof that this assumption is not coherent with the reality is that the Swedish government, to promote on-site production with PVs, owners have a tax credit for excess feed-in to grid set to expire in 2026 [10], that offered a feed-in premium of 0.6 SEK/kWh for small PV systems owners, showing that the assumption made overestimates the real selling price. A more realistic strategy would have prioritized self-consumption, which it was discovered while doing this project that IDA ICE is not the best to prioritize this, since PV production can be consumed only by facility. To do so, a realistic project would have integrated a battery storage system (BESS) to shift PV surplus to other times, reducing grid dependence and improving resilience. This approach would have probably led to a lower LEED total point.

Conclusion

Although the project achieved a LEED Platinum certification, the outcome should be interpreted considering the limitations discussed above. The high rating confirms strong performance within the boundaries of the LEED framework, but it doesn't fully resolve the mismatch between an international rating system (LEED) and the Swedish regulatory and energy context, nor does it compensate for the project's limited focus on demand reduction and passive strategies. The final result is the best seen as an indication of compliance with a specific certification logic, rather than a complete representation of sustainability performance. At the same time, the project has highlighted both strengths of the current design and clear opportunities for future improvement, such as prioritizing load reduction, circularity and self-consumption, before maximizing on-site generation.

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Appendix

Appendix 1- Water Consumption

Indoor water consumption

Baseline indoor water use is estimated using the following formula and data based on LEED, assuming an FTE (full-time equivalent) of 170:

EQUATION 1. Basic indoor water use reduction calculation

$$\text{Daily water use for each fixture type} = \text{Fixture flush or flow rate} \times \text{Duration of use} \times \text{Users} \times \text{Uses per person per day}$$

Figure A-1: Basic indoor water use reduction calculation from LEED

TABLE 8. Nonresidential default fixture uses					
Fixture type	Duration (sec)	Uses per day			
		Employees (FTE)	Visitors	Retail customers	Students
Water closet (female)	n/a	3	0.5	0.2	3
Water closet (male)	n/a	1	0.1	0.1	1
Urinal (female)	n/a	0	0	0	0
Urinal (male)	n/a	2	0.4	0.1	2
Public lavatory faucet	30	3	0.5	0.2	3
Shower	300	0.1	0	0	0
Kitchen sink	15	1	0	0	0

Figure A-2: Nonresidential default fixture uses from LEED

TABLE 1. Baseline water consumption of fixtures and fittings

Fixture or fitting	Baseline (IP units)	Baseline (SI units)
Toilet (water closet)*	1.6 gpf	6 lpf
Urinal*	1.0 gpf	3.8 lpf
Public lavatory (restroom) faucet	0.5 gpm at 60 psi all others except private applications	1.9 lpm at 415 kPa, all others except private applications
Private lavatory faucets	2.2 gpm at 60 psi	8.3 lpm at 415 kPa
Kitchen faucet (excluding faucets used exclusively for filling operations)	2.2 gpm at 60 psi	8.3 lpm at 415 kPa
Showerhead*	2.5 gpm at 80 psi per shower stall	9.5 lpm at 550 kPa per shower stall

*WaterSense label available for this product type

gpf = gallons per flush gpm = gallons per minute psi = pounds per square inch
 lpf = liters per flush lpm = liters per minute kPa = kilopascals

Figure A-3: baseline water consumption of fixtures and fittings from LEED

Considering also an outdoor water usage of 659 l/day in the baseline (LEED v5), reduced by 35% with the use of smart irrigation (to 428 l/day) [4][5].

Table A-1: Water usage calculation

Fixture type	Duration (s)	Use per day	Consumption (lpf or lpm)	Specific water consumption (per person per day)	Baseline water consumption (l per day)	Proposed water consumption (l per day)
Water closet female	-	3	6	18	1620	810
Water closet male	-	1	6	6	480	240
Urinal female	-	0	0	0	0	0
Urinal male	-	2	3.8	7.6	608	0
Public lavatory faucet	30	3	1.9	2.85	485	485
Shower	300	0.1	9.5	4.75	808	808
Outdoor water usage (550 m2)	-	-	-	-	659	428

The total water used in the baseline case (outdoor+indoor) accounts for 4,659 l/day. The proposed building, instead, has an average water consumption of 2,770 l/day.

Appendix 2- Overall Energy Performance

Baseline Building Consumptions and Costs:

Table A-2: Baseline Building Consumptions and Costs

Meter	Total, kWh	Cost, SEK
Purchased by facility (el)	19784	29676
Purchased by facility (dh)	96057	81649
Purchased by facility (dc)	20268	10134
Purchased by tenant (el)	142728	214092

Proposed Building Consumptions and Costs

Table A-3: Proposed Building Consumptions and Costs

Meter	Total, kWh	Cost, SEK
Exported by facility (el)	-89070	-133606
Purchased by facility (el)	6614	9922
Purchased by facility (dh)	95579	81242
Purchased by facility (dc)	20226	10113
Purchased by tenant (el)	142723	214085

Appendix 3- NPV Analysis for PV system

Cumulative NPV graph:

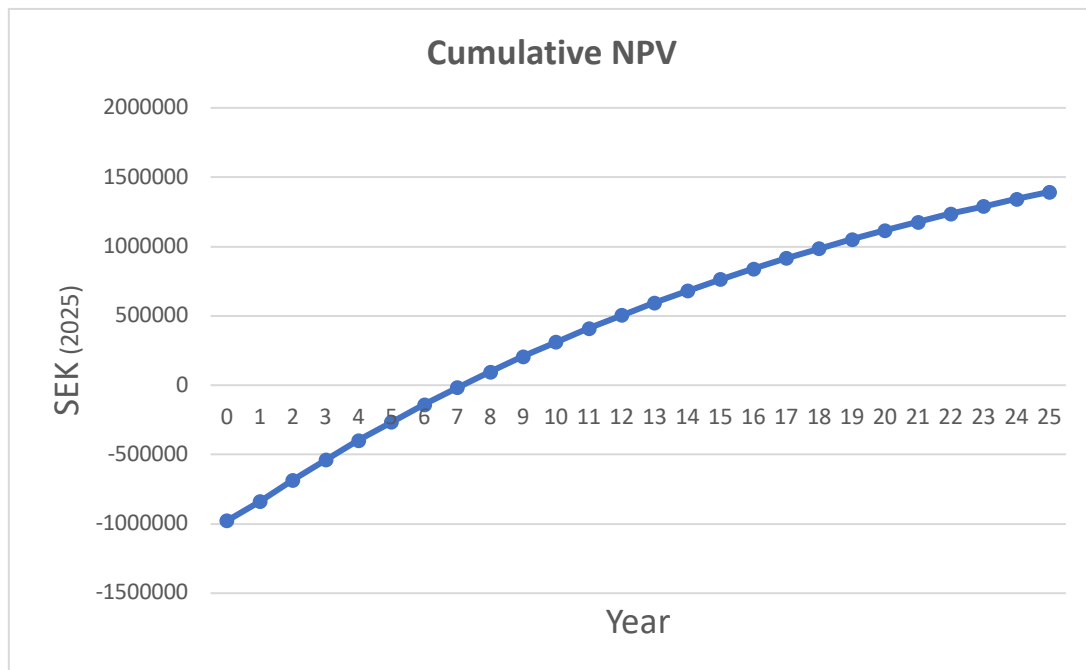


Figure A-4: PV system cumulative NPV graph

Calculations:

Table A-4: PV system calculations

Building	n° panels	kWp installed	CAPEX (SEK)	Savings per year (SEK)	Electricity Sold per Year (SEK)
Proposed	217	115.01	977,585	19,761	133,606

Yearly NPV calculations:

Table A-5: Yearly NPV calculations

Exp	977585	8050.7	8050.7	8050.7	8050.7	8050.7
revenue	0	153367	177972	177972	177972	177972
NPV	-977585	138397	154124	146784	139795	133138
Cumulative NPV	-977585	-839188	-685065	-538280	-398486	-265348
YEAR	0	1	2	3	4	5

Exp	8050.7	8050.7	8050.7	8050.7	8050.7	8050.7	8050.7	8050.7
revenue	177972	177972	177972	177972	177972	177972	177972	177972
NPV	126798	120760	115009	109533	104317	99349	94619	90113

Cumulative NPV	-138550	-17790	97219	206752	311069	410418	505037	595150
YEAR	6	7	8	9	10	11	12	13

Exp	8050.7	8050.7	8050.7	8050.7	8050.7	8050.7	8050.7	8050.7
revenue	177972	177972	177972	177972	177972	177972	177972	177972
NPV	85822	81735	77843	74136	70606	67244	64042	60992
Cumulative NPV	680972	762707	840550	914686	985292	1052535	1116577	1177569
YEAR	14	15	16	17	18	19	20	21

Exp	8050.7	8050.7	8050.7	8050.7
revenue	177972	177972	177972	177972
NPV	58088	55321	52687	50178
Cumulative NPV	1235656	1290978	1343665	1393843
YEAR	22	23	24	25

Appendix 4- Ventilation and IAQ Calculations

The outdoor air requirement for the main offices areas has been calculated according to the ASHRAE Ventilation Rate Procedure (*Buildingcalifornia*):

$$V_{bz} = R_p \cdot P_z + R_a \cdot A_z$$

For an office space, following Table 6-1 below:

- $R_p = 5 \text{ cfm/person} = 5 \cdot 1.699 \text{ m}^3/\text{h/person}$
- $R_a = 0.06 \text{ cfm/ft}^2 = 0.06 \cdot 1.699 / 10.764 = 1.10 \text{ m}^3/\text{h/m}^2$

With an estimated occupancy of 130 occupants (P_z) in a total area of 1405 m^2 (A_z). The total breathing-zone outdoor airflow is then $3858 \text{ m}^3/\text{h}$.

TABLE 6-1 MINIMUM VENTILATION RATES IN BREATHING ZONE
(This table is not valid in isolation; it must be used in conjunction with the accompanying notes.)

Occupancy Category	People Outdoor	Area Outdoor	Notes	Default Values		Air Class
	Air Rate	Air Rate		Occupant Density	Combined Outdoor	
	R_p	R_a		(see Note 4)	Air Rate (see Note 5)	
	cfm/person	cfm/ft ²		#/1000 ft ²	cfm/person	
Office Buildings						
Office space	5	0.06		5	17	1
Reception areas	5	0.06		30	7	1

Figure A-5: Minimum ventilation rates in breathing zones [11]

Appendix 5- Fanger's Index

Offices First - Summer (month and Week):

Date	Variables	
	PPD, Predicted Percentage of Dissatisfied, %	PMV, Predicted Mean Vote
2025-07-01	5.909	0.2807
2025-07-02	5.775	0.2788
2025-07-03	6.088	0.2908
2025-07-04	6.493	0.3121
2025-07-05	0.0	0.0
2025-07-06	0.0	0.0
2025-07-07	5.965	0.2832
2025-07-08	6.153	0.2961
2025-07-09	6.564	0.3169
2025-07-10	6.847	0.3309
2025-07-11	6.834	0.3304
2025-07-12	0.0	0.0
2025-07-13	0.0	0.0
2025-07-14	5.051	0.2342
2025-07-15	5.153	0.2428
2025-07-16	5.205	0.2458
2025-07-17	5.616	0.2688
2025-07-18	5.564	0.2653
2025-07-19	0.0	0.0
2025-07-20	0.0	0.0
2025-07-21	5.477	0.2544
2025-07-22	5.826	0.279
2025-07-23	5.988	0.2885
2025-07-24	6.511	0.3131
2025-07-25	6.27	0.3034
2025-07-26	0.0	0.0
2025-07-27	0.0	0.0
2025-07-28	5.906	0.281
2025-07-29	6.294	0.3031
2025-07-30	6.516	0.3147
2025-07-31	6.751	0.3259
mean	4.476	0.2142
mean*744.0 h	3330.1	159.4
min	0.0	0.0
max	6.847	0.3309

Figure A-6: Fanger Index

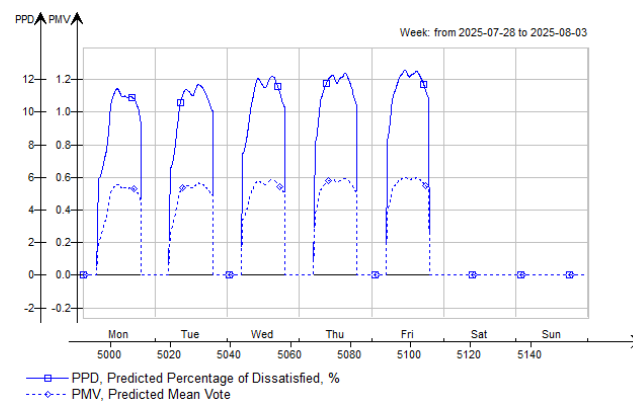


Figure A-7: PPD and PMV trends for Office first in summer

Offices First - Winter:

Date	Variables	
	PPD, Predicted Percentage of Dissatisfied, %	PMV, Predicted Mean Vote
2025-12-01	3.149	-0.05227
2025-12-02	3.165	-0.02342
2025-12-03	3.206	-0.04076
2025-12-04	3.208	-0.05954
2025-12-05	3.21	-0.03981
2025-12-06	0.0	0.0
2025-12-07	0.0	0.0
2025-12-08	3.193	-0.04541
2025-12-09	3.128	-0.03564
2025-12-10	3.127	-0.03887
2025-12-11	3.21	-0.04185
2025-12-12	3.115	-0.0212
2025-12-13	0.0	0.0
2025-12-14	0.0	0.0
2025-12-15	3.154	-0.0314
2025-12-16	3.152	-0.02543
2025-12-17	3.151	0.01651
2025-12-18	3.13	-0.01748
2025-12-19	3.186	-0.02129
2025-12-20	0.0	0.0
2025-12-21	0.0	0.0
2025-12-22	3.132	-0.03891
2025-12-23	3.11	-0.02
2025-12-24	0.0	0.0
2025-12-25	0.0	0.0
2025-12-26	0.0	0.0
2025-12-27	0.0	0.0
2025-12-28	0.0	0.0
2025-12-29	3.243	-0.07314
2025-12-30	3.15	-0.05105
2025-12-31	0.0	0.0
mean	1.939	-0.02132
mean*744.0 h	1442.9	-15.86
min	0.0	-0.07314
max	3.243	0.01651

Figure A-8: Fanger Index

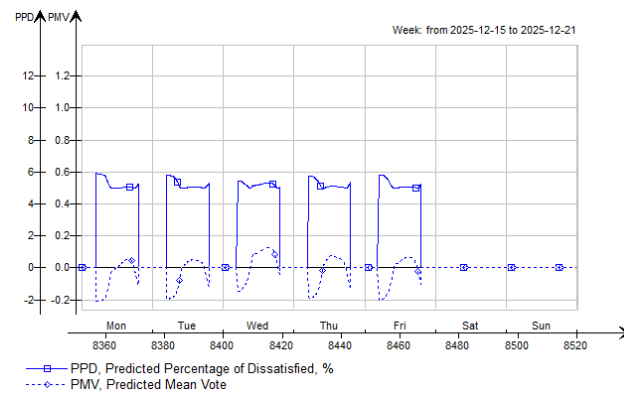


Figure A-9: PPD and PMV trends for Office first in winter

Gym - Summer:

Date	Variables	
	PPD, Predicted Percentage of Dissatisfied, %	PMV, Predicted Mean Vote
2025-07-01	17.54	0.6413
2025-07-02	17.49	0.6395
2025-07-03	18.25	0.6553
2025-07-04	18.51	0.6626
2025-07-05	0.0	0.0
2025-07-06	0.0	0.0
2025-07-07	17.07	0.6301
2025-07-08	17.78	0.6474
2025-07-09	18.14	0.6524
2025-07-10	18.65	0.6705
2025-07-11	18.49	0.6644
2025-07-12	0.0	0.0
2025-07-13	0.0	0.0
2025-07-14	14.93	0.5773
2025-07-15	15.96	0.6019
2025-07-16	16.25	0.6098
2025-07-17	17.35	0.6375
2025-07-18	17.16	0.6333
2025-07-19	0.0	0.0
2025-07-20	0.0	0.0
2025-07-21	15.88	0.5995
2025-07-22	17.47	0.6363
2025-07-23	17.88	0.6479
2025-07-24	18.88	0.6682
2025-07-25	18.33	0.6597
2025-07-26	0.0	0.0
2025-07-27	0.0	0.0
2025-07-28	17.08	0.6295
2025-07-29	18.13	0.6564
2025-07-30	18.32	0.6571
2025-07-31	18.47	0.66
mean	13.03	0.4754
mean*744.0 h	9696.5	353.7
min	0.0	0.0
max	18.88	0.6705

Figure A-10: Fanger Index

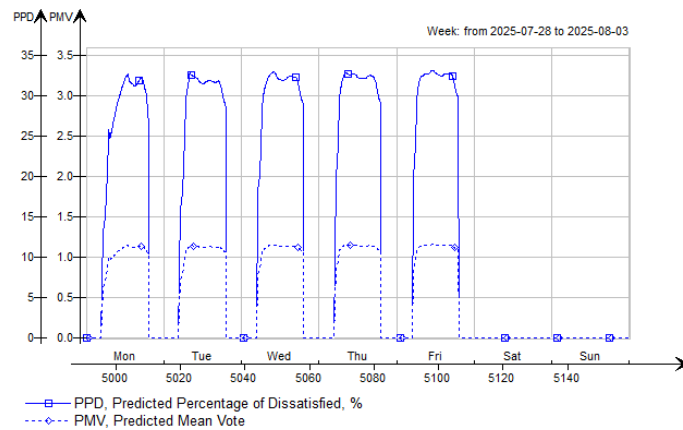


Figure A-11: PPD and PMV trends for Gym in summer

Gym - Winter:

Date	Variables	
	PPD, Predicted Percentage of Dissatisfied, %	PMV, Predicted Mean Vote
2025-12-01	7.193	0.3439
2025-12-02	8.169	0.3807
2025-12-03	7.84	0.3688
2025-12-04	7.407	0.3524
2025-12-05	8.115	0.3796
2025-12-06	0.0	0.0
2025-12-07	0.0	0.0
2025-12-08	7.338	0.35
2025-12-09	7.651	0.3619
2025-12-10	7.513	0.3569
2025-12-11	7.567	0.3593
2025-12-12	8.177	0.3811
2025-12-13	0.0	0.0
2025-12-14	0.0	0.0
2025-12-15	7.803	0.3669
2025-12-16	8.042	0.3768
2025-12-17	9.187	0.4172
2025-12-18	8.197	0.3819
2025-12-19	8.307	0.3867
2025-12-20	0.0	0.0
2025-12-21	0.0	0.0
2025-12-22	7.553	0.3579
2025-12-23	8.197	0.3823
2025-12-24	0.0	0.0
2025-12-25	0.0	0.0
2025-12-26	0.0	0.0
2025-12-27	0.0	0.0
2025-12-28	0.0	0.0
2025-12-29	6.768	0.3264
2025-12-30	7.436	0.3536
2025-12-31	0.0	0.0
mean	4.789	0.2253
mean*744.0 h	3563.0	167.6
min	0.0	0.0
max	9.187	0.4172

Figure A-12: Fanger Index

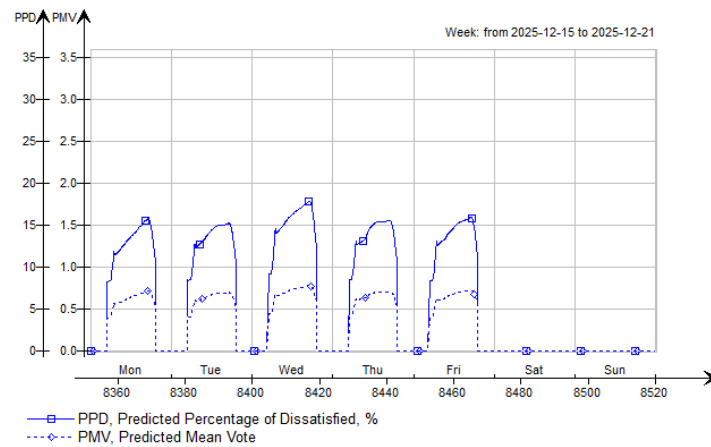


Figure A-13: PPD and PMV trends for Gym in summer

Appendix 6- Daylight Simulation Results

Table A-6: Daylight Simulation Results

Measuring plane	21/03/2025 09:00	21/03/2025 15:00	21/09/2025 09:00	21/09/2025 09:00
Offices First	1598	1548.9	1288.3	1796.7
Offices Second	659.2	1656.1	288.9	1953.1
Conference Room First 4	8186.8	513.4	6845.5	521.7
CEO Office	1553.4	5297	485.5	6430.1
CFO Office	401.8	1659.7	227.3	2990.6
Cafeteria Second	405.9	1046	347	621.1
Cafeteria First	359.9	867.8	300.7	534.1
CTO Office	5400.4	1688	4378.7	2436.2
Conference Room First 3	538.2	3798.4	306.6	6553.8
Conference Room First 2	458.4	4828.9	349.5	2970.3
Gym Ground	2051.9	380.6	1564.8	342.5
Entrance and Acceptance Ground	373.7	1626	170.6	2833.3
Common Area	2300.9	1485.8	930.1	1440.1